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  CONVEX-ROOF C_GME  -- METHOD-2 (Exponential Map)
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  Generated          : 2026-04-15  21:10
  System             : 1D Hubbard ring  (mode: uniform)

  SYSTEM PARAMETERS
  ------------------------------------------------
  Sites          L   = 4
  Electrons     Ne   = 4
  Hopping        t   = 1.0
  Interaction    U   = 4.0

  GROUND STATE
  ------------------------------------------------
  E0                 = -2.1027484835
  E1                 = -1.8064238518
  Gap  E1-E0         = 0.2963246316
  <S^2>              = 0.000000
  S                  = 0.0000
  <Sz>               = 0.000000
  Multiplicity       = 1.0
  Spin state         = Singlet (S=0)
  <D> total          = 0.287325
  <D> / site         = 0.071831

  ANALYSIS PARAMETERS
  ------------------------------------------------
  Clusters analysed  = 8
  Optimization Methods used       = 6
  Restarts / method  = 20

  OPTIMIZATION METHODS
  ------------------------------------------------
  1. L-BFGS-B
  2. Nelder-Mead
  3. Powell
  4. COBYLA
  5. Differential-Evo
  6. Simulated-Ann

  NOTATION
  ------------------------------------------------
  C_LB     = Lower bound  (paper coherence-witness, analytical)
  C_GME    = GME for pure rdm
  C_UB     = Upper bound  (eigendecomposition, analytical)
  C_GME_X  = Convex-roof estimate from optimiser X

  True C_GME satisfies:  C_LB <= C_GME <= C_UB

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  CLUSTER OVERVIEW  --  All Optimizers
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  | Cluster          |       Purity |         C_LB |        C_GME |         C_UB | C_GME_L-BFGS-B | C_GME_Nelder-M |   C_GME_Powell |   C_GME_COBYLA | C_GME_Diff-Evo |  C_GME_Sim-Ann | BestMethod   | Status |
  +------------------+--------------+--------------+--------------+--------------+----------------+----------------+----------------+----------------+----------------+----------------+--------------+--------+
  | (0, 1, 2)        |   0.46262004 |   0.00000000 |   0.00000000 |   0.71636717 |     0.50351431 | *   0.50351431 |     0.50351431 |     0.50351431 |     0.50351431 |     0.50351431 | Nelder-M     |    MIX |
  | (1, 2, 3)        |   0.46262004 |   0.00000000 |   0.00000000 |   0.71636717 |     0.50351431 | *   0.50351431 |     0.50351431 |     0.50351431 |     0.50351431 |     0.50351431 | Nelder-M     |    MIX |
  | (2, 3, 0)        |   0.46262004 |   0.00000000 |   0.00000000 |   0.71636717 |     0.50351431 | *   0.50351431 |     0.50351431 |     0.50351431 |     0.50351431 |     0.50351431 | Nelder-M     |    MIX |
  | (3, 0, 1)        |   0.46262004 |   0.00000000 |   0.00000000 |   0.71636717 |     0.50351431 | *   0.50351431 |     0.50351431 |     0.50351431 |     0.50351431 |     0.50351431 | Nelder-M     |    MIX |
  | (0, 1, 2, 3)     |   1.00000000 |   0.66666667 |   0.91287093 |   0.91287093 | *   0.91287093 |     0.91287093 |     0.91287093 |     0.91287093 |     0.91287093 |     0.91287093 | L-BFGS-B     |    GME |
  | (1, 2, 3, 0)     |   1.00000000 |   0.66666667 |   0.91287093 |   0.91287093 | *   0.91287093 |     0.91287093 |     0.91287093 |     0.91287093 |     0.91287093 |     0.91287093 | L-BFGS-B     |    GME |
  | (2, 3, 0, 1)     |   1.00000000 |   0.66666667 |   0.91287093 |   0.91287093 | *   0.91287093 |     0.91287093 |     0.91287093 |     0.91287093 |     0.91287093 |     0.91287093 | L-BFGS-B     |    GME |
  | (3, 0, 1, 2)     |   1.00000000 |   0.66666667 |   0.91287093 |   0.91287093 | *   0.91287093 |     0.91287093 |     0.91287093 |     0.91287093 |     0.91287093 |     0.91287093 | L-BFGS-B     |    GME |
  +------------------+--------------+--------------+--------------+--------------+----------------+----------------+----------------+----------------+----------------+----------------+--------------+--------+
  * = best (lowest) C_GME for that cluster

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  Optimizers SUMMARY  (aggregated across all clusters)
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  Optimizer            Mean C_GME    Std Dev         Best        Worst   Total(s)  Rank
  ------------------------------------------------------------------------------------
  Nelder-Mead          0.70819262   0.204678   0.50351431   0.91287093      11.48     1
  COBYLA               0.70819262   0.204678   0.50351431   0.91287093       9.56     2
  L-BFGS-B             0.70819262   0.204678   0.50351431   0.91287093       2.90     3
  Simulated-Ann        0.70819262   0.204678   0.50351431   0.91287093      12.08     4
  Powell               0.70819262   0.204678   0.50351431   0.91287093      13.30     5
  Differential-Evo     0.70819262   0.204678   0.50351431   0.91287093      67.47     6

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  TIMING ANALYSIS
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  Optimizer            Total(s)   Avg/cluster(s)  Fraction(%)  Rank(quality)
  ------------------------------------------------------------------------------
  Differential-Evo        67.47             8.43         57.8              6
  Powell                  13.30             1.66         11.4              5
  Simulated-Ann           12.08             1.51         10.3              4
  Nelder-Mead             11.48             1.43          9.8              1
  COBYLA                   9.56             1.19          8.2              2
  L-BFGS-B                 2.90             0.36          2.5              3
  ------------------------------------------------------------------------------
  TOTAL                  116.79


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================================================================================================================================================
  CLUSTER DETAIL  --  Sites (0, 1, 2)
================================================================================================================================================

  n (body order) = 3
  Purity Tr(r^2) = 0.4626200416
  Rank           = 4
  C_LB (lower)   = 0.0000000000
  C_GME   = 0.0000000000
  C_UB (upper)   = 0.7163671665
  Best C_GME (optimizer)     = 0.5035143074  [Nelder-Mead]
  Bounds : C_LB <= C_GME <= C_UB?     = YES

  Bipartition purities
  ------------------------------------------------------------
  A                    B                      Tr(rho_A^2)
  ------------------------------------------------------------
  (0,)                 (1, 2)                 0.5000000000
  (1,)                 (0, 2)                 0.5000000000
  (2,)                 (0, 1)                 0.5000000000

  Method results
  ------------------------------------------------------------------------------
  Method                       C_GME   Time(s)  Restarts   Evals        Spread
  ------------------------------------------------------------------------------
  L-BFGS-B             0.5035143074      0.70        20      20     5.62e-10
  Nelder-Mead          0.5035143074      2.90        20      10     1.44e-15
  Powell               0.5035143074      3.19        20       8     7.00e-11
  COBYLA               0.5035143074      2.51        20       6     3.55e-15
  Differential-Evo     0.5035143075     16.80        20       4     2.07e-09
  Simulated-Ann        0.5035143074      2.76        20       1     0.00e+00

  Pure-state decomposition  [Nelder-Mead]
  ------------------------------------------------
     #     weight p_i   C_GME(|psi_i>)      p_i*C_GME
  ------------------------------------------------
     1   0.2500000002     0.5035143077   0.1258785770
     2   0.2499999991     0.5035143064   0.1258785762
     3   0.2499999989     0.5035143061   0.1258785760
     4   0.2500000017     0.5035143094   0.1258785782
  ------------------------------------------------
   Sum                                  0.5035143074
  ( best C_GME = 0.5035143074)

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================================================================================================================================================
  CLUSTER DETAIL  --  Sites (1, 2, 3)
================================================================================================================================================

  n (body order) = 3
  Purity Tr(r^2) = 0.4626200416
  Rank           = 4
  C_LB (lower)   = 0.0000000000
  C_GME   = 0.0000000000
  C_UB (upper)   = 0.7163671665
  Best C_GME (optimizer)     = 0.5035143074  [Nelder-Mead]
  Bounds : C_LB <= C_GME <= C_UB?     = YES

  Bipartition purities
  ------------------------------------------------------------
  A                    B                      Tr(rho_A^2)
  ------------------------------------------------------------
  (0,)                 (1, 2)                 0.5000000000
  (1,)                 (0, 2)                 0.5000000000
  (2,)                 (0, 1)                 0.5000000000

  Method results
  ------------------------------------------------------------------------------
  Method                       C_GME   Time(s)  Restarts   Evals        Spread
  ------------------------------------------------------------------------------
  L-BFGS-B             0.5035143074      0.78        20      20     1.45e-10
  Nelder-Mead          0.5035143074      2.89        20      10     9.99e-16
  Powell               0.5035143074      3.73        20       8     6.39e-11
  COBYLA               0.5035143074      2.44        20       6     3.31e-14
  Differential-Evo     0.5035143074     16.79        20       4     9.97e-10
  Simulated-Ann        0.5035143074      3.25        20       1     0.00e+00

  Pure-state decomposition  [Nelder-Mead]
  ------------------------------------------------------
     #     weight p_i   C_GME(|psi_i>)      p_i*C_GME
  ------------------------------------------------------
     1   0.2500000008     0.5035143083   0.1258785775
     2   0.2499999981     0.5035143051   0.1258785753
     3   0.2499999999     0.5035143073   0.1258785768
     4   0.2500000012     0.5035143088   0.1258785778
  ------------------------------------------------------
   Sum                                  0.5035143074
  ( best C_GME = 0.5035143074)

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================================================================================================================================================
  CLUSTER DETAIL  --  Sites (2, 3, 0)
================================================================================================================================================

  n (body order) = 3
  Purity Tr(r^2) = 0.4626200416
  Rank           = 4
  C_LB (lower)   = 0.0000000000
  C_GME   = 0.0000000000
  C_UB (upper)   = 0.7163671679
  Best C_GME (optimizer)     = 0.5035143074  [Nelder-Mead]
  Bounds : C_LB <= C_GME <= C_UB?     = YES

  Bipartition purities
  ------------------------------------------------------------
  A                    B                      Tr(rho_A^2)
  ------------------------------------------------------------
  (0,)                 (1, 2)                 0.5000000000
  (1,)                 (0, 2)                 0.5000000000
  (2,)                 (0, 1)                 0.5000000000

  Method results
  ------------------------------------------------------------------------------
  Method                       C_GME   Time(s)  Restarts   Evals        Spread
  ------------------------------------------------------------------------------
  L-BFGS-B             0.5035143074      0.74        20      20     8.12e-10
  Nelder-Mead          0.5035143074      2.85        20      10     5.55e-16
  Powell               0.5035143074      2.97        20       8     4.65e-11
  COBYLA               0.5035143074      2.17        20       6     3.33e-15
  Differential-Evo     0.5035143074     17.05        20       4     9.06e-10
  Simulated-Ann        0.5035143074      3.13        20       1     0.00e+00

  Pure-state decomposition  [Nelder-Mead]
  ------------------------------------------------
     #     weight p_i   C_GME(|psi_i>)      p_i*C_GME
  ------------------------------------------------
     1   0.2500000005     0.5035143080   0.1258785772
     2   0.2500000044     0.5035143125   0.1258785803
     3   0.2499999996     0.5035143069   0.1258785765
     4   0.2499999955     0.5035143022   0.1258785733
  ------------------------------------------------
   Sum                                  0.5035143074
  ( best C_GME = 0.5035143074)

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================================================================================================================================================
  CLUSTER DETAIL  --  Sites (3, 0, 1)
================================================================================================================================================

  n (body order) = 3
  Purity Tr(r^2) = 0.4626200416
  Rank           = 4
  C_LB (lower)   = 0.0000000000
  C_GME   = 0.0000000000
  C_UB (upper)   = 0.7163671668
  Best C_GME (optimizer)     = 0.5035143074  [Nelder-Mead]
  Bounds : C_LB <= C_GME <= C_UB?     = YES

  Bipartition purities
  ------------------------------------------------------------
  A                    B                      Tr(rho_A^2)
  ------------------------------------------------------------
  (0,)                 (1, 2)                 0.5000000000
  (1,)                 (0, 2)                 0.5000000000
  (2,)                 (0, 1)                 0.5000000000

  Method results
  ------------------------------------------------------------------------------
  Method                       C_GME   Time(s)  Restarts   Evals        Spread
  ------------------------------------------------------------------------------
  L-BFGS-B             0.5035143074      0.68        20      20     3.53e-10
  Nelder-Mead          0.5035143074      2.83        20      10     8.88e-16
  Powell               0.5035143074      3.42        20       8     8.61e-11
  COBYLA               0.5035143074      2.45        20       6     5.44e-15
  Differential-Evo     0.5035143074     16.82        20       4     1.00e-09
  Simulated-Ann        0.5035143074      2.94        20       1     0.00e+00

  Pure-state decomposition  [Nelder-Mead]
  ------------------------------------------------
     #     weight p_i   C_GME(|psi_i>)      p_i*C_GME
  ------------------------------------------------
     1   0.2500000005     0.5035143080   0.1258785772
     2   0.2500000013     0.5035143089   0.1258785779
     3   0.2499999975     0.5035143044   0.1258785748
     4   0.2500000007     0.5035143082   0.1258785774
  ------------------------------------------------
   Sum                                  0.5035143074
  ( best C_GME = 0.5035143074)

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```
================================================================================================================================================
  CLUSTER DETAIL  --  Sites (0, 1, 2, 3)
================================================================================================================================================

  n (body order) = 4
  Purity Tr(r^2) = 1.0000000000
  Rank           = 1
  C_LB (lower)   = 0.6666666667
  C_GME   = 0.9128709292
  C_UB (upper)   = 0.9128709292
  Best C_GME (optimizer)     = 0.9128709292  [L-BFGS-B]
  Bounds : C_LB <= C_GME <= C_UB?     = YES

  Bipartition purities
  ------------------------------------------------------------
  A                    B                      Tr(rho_A^2)
  ------------------------------------------------------------
  (0,)                 (1, 2, 3)              0.5000000000
  (1,)                 (0, 2, 3)              0.5000000000
  (2,)                 (0, 1, 3)              0.5000000000
  (3,)                 (0, 1, 2)              0.5000000000
  (0, 1)               (2, 3)                 0.5833333333
  (0, 2)               (1, 3)                 0.3333333333
  (1, 2)               (0, 3)                 0.3333333333

  Method results
  ------------------------------------------------------------------------------
  Method                       C_GME   Time(s)  Restarts   Evals       Spread
  ------------------------------------------------------------------------------
  L-BFGS-B             0.9128709292      0.00         0       0     0.00e+00
  Nelder-Mead          0.9128709292      0.00         0       0     0.00e+00
  Powell               0.9128709292      0.00         0       0     0.00e+00
  COBYLA               0.9128709292      0.00         0       0     0.00e+00
  Differential-Evo     0.9128709292      0.00         0       0     0.00e+00
  Simulated-Ann        0.9128709292      0.00         0       0     0.00e+00

  Pure-state decomposition  [L-BFGS-B]
  ------------------------------------------------------------
     #     weight p_i   C_GME(|psi_i>)      p_i*C_GME
  ------------------------------------------------------------
     1   1.0000000000     0.9128709292   0.9128709292
  ------------------------------------------------------------
   Sum                                   0.9128709292
  ( best C_GME = 0.9128709292)

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```
================================================================================================================================================
  CLUSTER DETAIL  --  Sites (1, 2, 3, 0)
================================================================================================================================================

  n (body order) = 4
  Purity Tr(r^2) = 1.0000000000
  Rank           = 1
  C_LB (lower)   = 0.6666666667
  C_GME   = 0.9128709292
  C_UB (upper)   = 0.9128709292
  Best C_GME (optimizer)     = 0.9128709292  [L-BFGS-B]
  Bounds : C_LB <= C_GME <= C_UB?     = YES

  Bipartition purities
  ------------------------------------------------------------
  A                    B                      Tr(rho_A^2)
  ------------------------------------------------------------
  (0,)                 (1, 2, 3)              0.5000000000
  (1,)                 (0, 2, 3)              0.5000000000
  (2,)                 (0, 1, 3)              0.5000000000
  (3,)                 (0, 1, 2)              0.5000000000
  (0, 1)               (2, 3)                 0.5833333333
  (0, 2)               (1, 3)                 0.3333333333
  (1, 2)               (0, 3)                 0.3333333333

  Method results
  ------------------------------------------------------------------------------
  Method                      C_GME   Time(s)  Restarts   Evals       Spread
  ------------------------------------------------------------------------
  L-BFGS-B             0.9128709292      0.00         0       0     0.00e+00
  Nelder-Mead          0.9128709292      0.00         0       0     0.00e+00
  Powell               0.9128709292      0.00         0       0     0.00e+00
  COBYLA               0.9128709292      0.00         0       0     0.00e+00
  Differential-Evo     0.9128709292      0.00         0       0     0.00e+00
  Simulated-Ann        0.9128709292      0.00         0       0     0.00e+00

  Pure-state decomposition  [L-BFGS-B]
  ------------------------------------------------------------
     #     weight p_i   C_GME(|psi_i>)      p_i*C_GME
  ------------------------------------------------------
     1   1.0000000000     0.9128709292   0.9128709292
  ------------------------------------------------------
   Sum                                   0.9128709292
  ( best C_GME = 0.9128709292)

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```
================================================================================================================================================
  CLUSTER DETAIL  --  Sites (2, 3, 0, 1)
================================================================================================================================================

  n (body order) = 4
  Purity Tr(r^2) = 1.0000000000
  Rank           = 1
  C_LB (lower)   = 0.6666666667
  C_GME   = 0.9128709292
  C_UB (upper)   = 0.9128709292
  Best C_GME (optimizer)     = 0.9128709292  [L-BFGS-B]
  Bounds : C_LB <= C_GME <= C_UB?     = YES

  Bipartition purities
  ------------------------------------------------------------
  A                    B                      Tr(rho_A^2)
  ------------------------------------------------------------
  (0,)                 (1, 2, 3)              0.5000000000
  (1,)                 (0, 2, 3)              0.5000000000
  (2,)                 (0, 1, 3)              0.5000000000
  (3,)                 (0, 1, 2)              0.5000000000
  (0, 1)               (2, 3)                 0.5833333333
  (0, 2)               (1, 3)                 0.3333333333
  (1, 2)               (0, 3)                 0.3333333333

  Method results
  ------------------------------------------------------------------------------
  Method                      C_GME   Time(s)  Restarts   Evals       Spread
  ------------------------------------------------------------------------
  L-BFGS-B             0.9128709292      0.00         0       0     0.00e+00
  Nelder-Mead          0.9128709292      0.00         0       0     0.00e+00
  Powell               0.9128709292      0.00         0       0     0.00e+00
  COBYLA               0.9128709292      0.00         0       0     0.00e+00
  Differential-Evo     0.9128709292      0.00         0       0     0.00e+00
  Simulated-Ann        0.9128709292      0.00         0       0     0.00e+00

  Pure-state decomposition  [L-BFGS-B]
  ------------------------------------------------------------
     #     weight p_i   C_GME(|psi_i>)      p_i*C_GME
  ------------------------------------------------------
     1   1.0000000000     0.9128709292   0.9128709292
  ------------------------------------------------------
   Sum                                   0.9128709292
  ( best C_GME = 0.9128709292)

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```
================================================================================================================================================
  CLUSTER DETAIL  --  Sites (3, 0, 1, 2)
================================================================================================================================================

  n (body order) = 4
  Purity Tr(r^2) = 1.0000000000
  Rank           = 1
  C_LB (lower)   = 0.6666666667
  C_GME   = 0.9128709292
  C_UB (upper)   = 0.9128709292
  Best C_GME (optimizer)     = 0.9128709292  [L-BFGS-B]
  Bounds : C_LB <= C_GME <= C_UB?     = YES

  Bipartition purities
  ------------------------------------------------------------
  A                    B                      Tr(rho_A^2)
  ------------------------------------------------------------
  (0,)                 (1, 2, 3)              0.5000000000
  (1,)                 (0, 2, 3)              0.5000000000
  (2,)                 (0, 1, 3)              0.5000000000
  (3,)                 (0, 1, 2)              0.5000000000
  (0, 1)               (2, 3)                 0.5833333333
  (0, 2)               (1, 3)                 0.3333333333
  (1, 2)               (0, 3)                 0.3333333333

  Method results
  ------------------------------------------------------------------------------
  Method                       C_GME   Time(s)  Restarts   Evals       Spread
  ------------------------------------------------------------------------------
  L-BFGS-B             0.9128709292      0.00         0       0     0.00e+00
  Nelder-Mead          0.9128709292      0.00         0       0     0.00e+00
  Powell               0.9128709292      0.00         0       0     0.00e+00
  COBYLA               0.9128709292      0.00         0       0     0.00e+00
  Differential-Evo     0.9128709292      0.00         0       0     0.00e+00
  Simulated-Ann        0.9128709292      0.00         0       0     0.00e+00

  Pure-state decomposition  [L-BFGS-B]
  ------------------------------------------------------------
     #     weight p_i   C_GME(|psi_i>)      p_i*C_GME
  ------------------------------------------------------------
     1   1.0000000000     0.9128709292   0.9128709292
  ------------------------------------------------------------
   Sum                                   0.9128709292
  ( best C_GME = 0.9128709292)

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